inverse matricies and Gaussian elimination continued

1) What is a function that we know is invertible? I

$(I \circ I) = I$

2) Try to apply Gassian elimination steps to obtain I

EX:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{\frac{1}{3}r_2} \begin{bmatrix} 1 & 2 \\ 1 & 4/3 \end{bmatrix}$$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right] \xrightarrow{\frac{1}{3}r_2} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 1 & 4/3 & 0 & 1/3 \end{array}\right] \xrightarrow{r_2 - r_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -\frac{2}{3} & -1 & \frac{1}{3} \end{array}\right] \xrightarrow{-\frac{3}{2}r_2} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

$r_1 - 2r_2$

$$\left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

check

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix} = \begin{bmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Ex: $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{r_2 - r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$ not invertible

lemma: column (or row) of 0's means non invertible

lemma: 2 identical rows means non invertible

Invertibility test 1: row reducible to I is equivalent to invertible

why do we care about invertibility?

$$\begin{aligned} 2x + 3y &= 7 \\ -x + y &= 4 \end{aligned} \implies \underbrace{\begin{bmatrix} 2 & 3 \\ -1 & 1 \end{bmatrix}}_{A}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix} \qquad A\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \end{bmatrix}$$

1) if we know A has an inverse then

$$A\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}7\\4\end{bmatrix} \Rightarrow A^{-1}A\begin{bmatrix}x\\y\end{bmatrix}=A^{-1}\begin{bmatrix}7\\4\end{bmatrix}$$

$$I\begin{bmatrix}x\\y\end{bmatrix}=A^{-1}\begin{bmatrix}7\\4\end{bmatrix}$$

$$\begin{bmatrix}x\\y\end{bmatrix}=A^{-1}\begin{bmatrix}7\\4\end{bmatrix}$$

So our original system has exactly 1 solution

is $(7,4)\in \text{Span}\left((2,-1),(3,1)\right)$?

$$(7,4)=x(2,-1)+y(3,1)$$

2) if we can compute A^{-1} then we can compute $\begin{bmatrix}x\\y\end{bmatrix}$

<u>EX</u>: $\left.\begin{matrix}(1,1,0)\\(1,0,1)\\(0,1,1)\end{matrix}\right\}$ are linearly independent iff $\begin{bmatrix}1&1&0\\1&0&1\\0&1&1\end{bmatrix}$ has an inverse

linear independence happens when $0=x(1,1,0)+y(1,0,1)+z(0,1,1)$ has exactly 1 sol

$$\begin{bmatrix}1&1&0\\1&0&1\\0&1&1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc}1&1&0&1&0&0\\1&0&1&0&1&0\\0&1&1&0&0&1\end{array}\right]\xrightarrow{r_2-r_1}\left[\begin{array}{ccc|ccc}1&1&0&1&0&0\\0&-1&1&-1&1&0\\0&1&1&0&0&1\end{array}\right]\xrightarrow{r_2-r_3}\left[\begin{array}{ccc|ccc}1&1&0&1&0&0\\0&-2&0&-1&1&-1\\0&1&1&0&0&1\end{array}\right]$$

$$\xrightarrow{-\frac{1}{2}r_2}\left[\begin{array}{ccc|ccc}1&1&0&1&0&0\\0&1&0&\frac{1}{2}&-\frac{1}{2}&\frac{1}{2}\\0&1&1&0&0&1\end{array}\right]\xrightarrow{r_1-r_2,\ r_3-r_2}\left[\begin{array}{ccc|ccc}1&0&0&\frac{1}{2}&\frac{1}{2}&-\frac{1}{2}\\0&1&0&\frac{1}{2}&-\frac{1}{2}&\frac{1}{2}\\0&0&1&-\frac{1}{2}&\frac{1}{2}&\frac{1}{2}\end{array}\right]$$

$\left.\begin{matrix}\frac{1}{2}&\frac{1}{2}&-\frac{1}{2}\\\frac{1}{2}&-\frac{1}{2}&\frac{1}{2}\\-\frac{1}{2}&\frac{1}{2}&\frac{1}{2}\end{matrix}\right\}$ inverse